

Q.1

Show that $\frac{d}{dz}(\operatorname{arcsinh}(z)) = \frac{1}{\sqrt{1+z^2}}$

Proof: -

$$\text{let } w = \operatorname{arcsinh} z, \quad \frac{dw}{dz} = ?$$

suppose

$$z = \sinh w$$

$$\frac{dz}{dw} = \cosh w \cdot \frac{dw}{dz}$$

$$1 = \cosh w \cdot \frac{dw}{dz}$$

$$\frac{dw}{dz} = \frac{1}{\cosh w} \quad \rightarrow \textcircled{1}$$

$$\therefore \cosh^2 w - \sinh^2 w = 1$$

M.M

$$\cosh^2 w = 1 + \sinh^2 w$$

$$\sqrt{\cosh^2 w} = \sqrt{1 + \sinh^2 w}$$

$$\cosh w = \sqrt{1 + \sinh^2 w}$$

by $\textcircled{1}$

$$\frac{dw}{dz} = \frac{1}{\sqrt{1 + \sinh^2 w}}$$

$$\text{let } z = \sinh w$$

$$z^2 = \sinh^2 w$$

$$\frac{dw}{dz} = \frac{1}{\sqrt{1 + z^2}}$$

$$\frac{d}{dz}(\operatorname{arcsinh}(z)) = \frac{1}{\sqrt{1+z^2}}$$

Q.2

Evaluate the integral

$$\int_0^{\pi/6} e^{i2t} dt$$

Sol:-

M.M

$$\int_0^{\pi/6} e^{i2t} dt$$

$$= \int_0^{\pi/6} (\cos 2t + i \sin 2t) dt$$

$$= \int_0^{\pi/6} \cos 2t dt + i \int_0^{\pi/6} \sin 2t dt$$

$$= \left. \frac{\sin 2t}{2} \right|_0^{\pi/6} + i \left(\left. -\frac{\cos 2t}{2} \right|_0^{\pi/6} \right)$$

$$= \frac{1}{2} \left(\sin 2\left(\frac{\pi}{6}\right) - 0 \right) + \frac{i}{2} \left(\cos 2\left(\frac{\pi}{6}\right) - \cos 0 \right)$$

$$= \frac{1}{2} \left(\sin \frac{\pi}{3} \right) + \frac{i}{2} \left(\cos \frac{\pi}{3} - 1 \right)$$

$$= \frac{1}{2} \left(\frac{\sqrt{3}}{2} \right) + \frac{i}{2} \left(\frac{1}{2} - 1 \right)$$

$$= \frac{\sqrt{3}}{4} + \frac{i}{2} \left(\frac{1-2}{2} \right)$$

$$= \frac{\sqrt{3}}{4} + \frac{i}{2} \left(-\frac{1}{2} \right)$$

$$= \frac{\sqrt{3}}{4} - \frac{i}{4} = \frac{\sqrt{3} - i}{4} \quad \checkmark$$

Q.3

Show that $\arcsin(z) = -i \log(iz + (1-z^2)^{1/2})$

Sol.:

$$\arcsin(z) = -i \log(iz + (1-z^2)^{1/2})$$

let

$$w = \sin^{-1} z \quad \text{when } z = \sin w$$

$$\text{when } w = \frac{e^{iw} - e^{-iw}}{2i}$$

put this eq. in the form

$$(e^{iw})^2 - 2iz(e^{iw}) - 1 = 0$$

by solving e^{iw} we get

$$e^{iw} = iz + (1-z^2)^{1/2} \rightarrow (1)$$

Here $(1-z^2)^{1/2}$ is a double-valued function of z . taking log both side of (1)

$$\log e^{iw} = \log(iz + (1-z^2)^{1/2})$$

$$iw = \log(iz + (1-z^2)^{1/2})$$

Multiply with i both side

$$i^2 w = i \log(iz + (1-z^2)^{1/2})$$

$$-w = i \log(iz + (1-z^2)^{1/2})$$

$$w = -i \log(iz + (1-z^2)^{1/2})$$

$$\therefore w = \sin^{-1} z$$

$$\sin^{-1} z = -i \log(iz + (1-z^2)^{1/2})$$

$$\arcsin(z) = -i \log(iz + (1-z^2)^{1/2})$$

Q.4

Compute $V_p[f]$ for $f = x^2yz$ with
 $p = (1, 1, 0)$ and $V = (1, 0, -3)$

Sol:-

$$f = x^2yz$$

$$\frac{\partial f}{\partial x} = 2xyz$$

$$\frac{\partial f}{\partial y} = x^2z$$

$$\frac{\partial f}{\partial z} = x^2y$$

at point $P(1, 1, 0)$

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0, \quad \frac{\partial f}{\partial z} = 1$$

So,

$$V_p[f] = \frac{\partial f}{\partial x} \cdot V_1 + \frac{\partial f}{\partial y} \cdot V_2 + \frac{\partial f}{\partial z} \cdot V_3$$

$$= (0)(1) + (0)(0) + (1)(-3)$$

$$= -3$$

Q.5

Write $-3i^4$ in the form $a+ib$

$$a+ib = -3i^4$$

$$= -3(i^2)^2$$

$$= -3(-1)^2$$

$$= -3 + 0i$$

Q.6

Show that $\text{Log}(1+i)^2 = 2 \log(1+i)$
and $\log(-1+i)^2 \neq 2 \log(-1+i)$

Ans:-

$$\textcircled{1} \text{Log}(1+i)^2 = 2 \log(1+i)$$

Let

$$\begin{aligned} (1+i) &= \sqrt{2} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \\ &= \sqrt{2} (e^{i\pi/4}) \\ &= \sqrt{2} e^{i\pi/4} \end{aligned}$$

$$(1+i)^2 = (\sqrt{2} e^{i\pi/4})^2 = 2 e^{i\pi/2}$$

therefore $\text{Arg}(1+i) = \frac{\pi}{4}$ and $\text{Arg}(1+i)^2 = \frac{\pi}{2}$

$$\log(1+i)^2 = \ln|(1+i)^2| + i \text{Arg}(1+i)^2$$

$$= \ln|2e^{i\pi/2}| + i \frac{\pi}{2}$$

$$= \ln|2(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})| + i \frac{\pi}{2}$$

$$= \ln|2| + i \frac{\pi}{2}$$

$$= 2 \left(\frac{1}{2} \ln|2| + i \frac{\pi}{4} \right)$$

$$= 2 \left(\ln\sqrt{2} + i \frac{\pi}{4} \right)$$

$$= 2 \log(1+i)$$

Q.6-Part-2

$$\log(-1+i)^2 \neq 2\log(-1+i)$$

L.H.S

$$-1+i = \sqrt{2} \left(-\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) = \sqrt{2} e^{i \frac{3\pi}{4}}$$

$$(-1+i)^2 = \left(\sqrt{2} e^{i \frac{3\pi}{4}} \right)^2 = 2 e^{i \frac{3\pi}{2}}$$

$$\text{therefore } \text{Arg}(-1+i) = \frac{3\pi}{4} \text{ and } \text{Arg}(-1+i)^2 = \frac{3\pi}{2}$$

$$\log(-1+i)^2 = \ln|(-1+i)^2| + i \text{Arg}(-1+i)^2$$

$$= \ln|2 e^{i \frac{3\pi}{2}}| + i \frac{3\pi}{2}$$

$$= \ln|2(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2})| + i \frac{3\pi}{2}$$

$$= \ln|2| + i \frac{3\pi}{2} \neq 2\log(-1+i)$$

thus,

$$\log(-1+i)^2 \neq 2\log(-1+i)$$